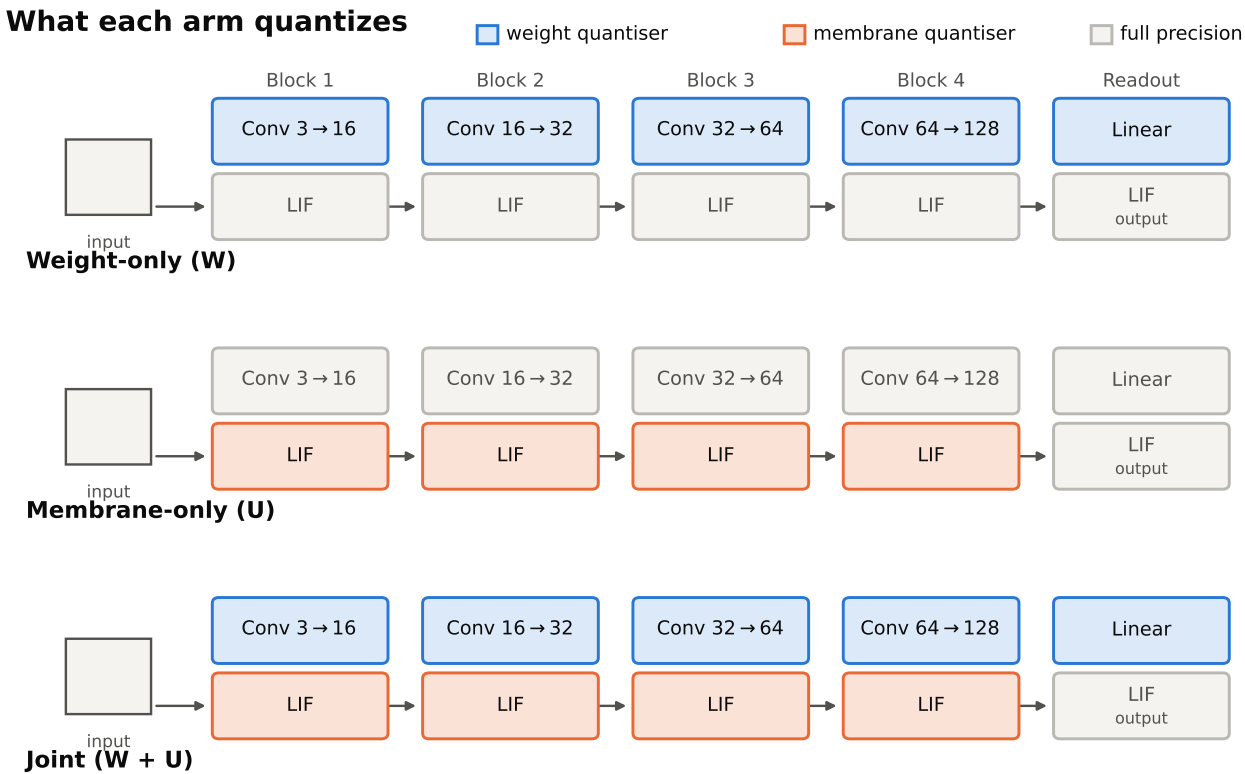
What each arm quantizes
weight quantiser
membrane quantiser
full precision
Block 1
Block 2
Block 3
Block 4
Readout
Conv 3→16
Conv 16→32
Conv 32→64
Conv 64→128
Linear
LIF
LIF
LIF
LIF
LIF output
input
Weight-only (W)
Conv 3→16
Conv 16→32
Conv 32→64
Conv 64→128
Linear
LIF
LIF
LIF
LIF
LIF output
input
Membrane-only (U)
Conv 3→16
Conv 16→32
Conv 32→64
Conv 64→128
Linear
LIF
LIF
LIF
LIF
LIF output
input
Joint (W + U)